# Urban Building-Material RF Model

Physics-grounded reflection, transmission, penetration, scattering, and O2I modelling for urban RF simulation

| VERSION | PRIMARY STANDARDS | INTENDED RANGE |
|---|---|---|
| 1.3.0 | ITU-R P.2040 · 3GPP TR 38.901 | Sub-6 GHz through mmWave |

## Purpose and scope

A production-oriented reference for RF reflection, transmission, penetration, absorption, and diffuse scattering from urban building materials. It supports deterministic ray tracing, hybrid RF digital twins, and 3GPP-aligned outdoor-to-indoor (O2I) simulation from sub-6 GHz through mmWave.

> Core principle: retain complex field amplitude and phase for deterministic paths. Convert to power only after coherent receiver-field summation.

| Tier | Representation | Use case | Limitation |
|---|---|---|---|
| 0 | Opaque blocker / fixed loss | Rapid outdoor coverage | No path-specific reflections or O2I detail |
| 1 | Single-interface Fresnel coefficient | Facade reflection rays | No finite thickness or layer structure |
| 2 | Finite lossy slab | Concrete, brick, plain glass, timber | Misses coatings and detailed reinforcement |
| 3 | Multilayer transfer matrix | Curtain walls, glazing, insulated facades | Needs layer order, thickness, and material data |
| 4 | Explicit rebar/roughness/apertures | Critical mmWave analysis | High compute and measurement burden |
| 5 | Statistical O2I model | Network/system simulations | Not facade-specific |

Preferred city-scale architecture: deterministic outdoor geometry and dominant rays; material archetypes or stacks for facades; calibrated uncertainty for unobserved construction variation.

## Energy partition and material state

$$P_i = P_{spec} + P_{diff} + P_t + P_a$$

Incident energy partitions into coherent specular reflection, diffuse scatter, transmitted power, and absorbed power. Ideal smooth lossless interfaces have R + T = 1; real facades require R + T + A + S = 1.

$$\tilde{\varepsilon}_r(f) = \varepsilon'_r(f) - j[\varepsilon''_r(f) + \sigma(f)/(\omega\varepsilon_0)],\ \omega = 2\pi f$$

$$\gamma = \alpha + j\beta = j\omega \sqrt{\{ \mu [ \varepsilon - j\sigma/\omega ] \}}$$

$$E(z) = E_0 e^{(-\gamma z)} = E_0 e^{(-\alpha z)} e^{(-j\beta z)},\ L_{bulk} = 8.686\,\alpha d\ [dB]$$

Bulk attenuation is not total wall penetration loss. Include coupling at entry and exit, layer structure, internal reflection, roughness, reinforcement, and coatings.

# Local geometry and polarization

$$\cos\theta_i = -\hat{k}_i \cdot \hat{n}$$

$$\hat{k}_r = \hat{k}_i - 2(\hat{k}_i \cdot \hat{n})\hat{n}$$

$$\hat{s} = (\hat{n} \times \hat{k}_i) / ||\hat{n} \times \hat{k}_i||, \hat{p}_i = \hat{s} \times \hat{k}_i$$

$$E_{TE} = E_i \cdot \hat{s}, E_{TM} = E_i \cdot \hat{p}_i$$

Calculate local incidence against the actual surface normal. Transform global antenna polarization into local TE/TM components at every hit point; a global ±45° antenna basis is not the Fresnel basis.

# Fresnel reflection and transmission

$$\Gamma_{TE} = (\tilde{\eta}_2 \cos\theta_i - \tilde{\eta}_1 \cos\theta_t) / (\tilde{\eta}_2 \cos\theta_i + \tilde{\eta}_1 \cos\theta_t)$$

$$\Gamma_{TM} = (\tilde{\eta}_2 \cos\theta_t - \tilde{\eta}_1 \cos\theta_i) / (\tilde{\eta}_2 \cos\theta_t + \tilde{\eta}_1 \cos\theta_i)$$

$$\tilde{n}_1 \sin\theta_i = \tilde{n}_2 \sin\theta_t, R = |\Gamma|^2, L_{refl} = -10 \log_{10}(R)$$

At normal incidence: $\Gamma = (\tilde{n}_1 - \tilde{n}_2)/(\tilde{n}_1 + \tilde{n}_2)$

Evaluate TE and TM separately with complex material properties and apply coefficients to complex electric fields. For air to an ideal dielectric with $\varepsilon_r = 4$, $\Gamma = -1/3$ and front-face reflected power is 1/9, about 11.1%. This does not represent the total response of a finite wall.

# Finite slabs and multilayer facades

$$k_{z,1} = k_0 \sqrt{(\tilde{\varepsilon}_{r,1} - \sin^2\theta_i)}, k_0 = 2\pi f/c$$

$$T_{slab} = [t_{01} t_{12} e^{(-jk_{z,1}d)}] / [1 + r_{01} r_{12} e^{(-j2k_{z,1}d)}]$$

$$R_{slab} = [r_{01} + r_{12} e^{(-j2k_{z,1}d)}] / [1 + r_{01} r_{12} e^{(-j2k_{z,1}d)}]$$

For air–slab–air: $L_{pen} = -10 \log_{10}(|T_{slab}|^2)$

A finite slab includes entry reflection, propagation and absorption, exit reflection, and repeated internal reflections. Retain complex transmission for narrowband, phase-sensitive, and deterministic ray-channel work.

$$M_i = [[\cos(k_{z,i}d_i), j \sin(k_{z,i}d_i)/Y_i], [jY_i \sin(k_{z,i}d_i), \cos(k_{z,i}d_i)]]$$

$$M = \Pi_i M_i = [[A,B],[C,D]]$$

$$r = (Y_{0A} + Y_0 Y_{sB} - C - Y_{sD})/(Y_{0A} + Y_0 Y_{sB} + C + Y_{sD})$$

$$t = 2Y_0/(Y_{0A} + Y_0 Y_{sB} + C + Y_{sD}), T = \mathrm{Re}(Y_s)/\mathrm{Re}(Y_0) \cdot |t|^2$$

Use transfer matrices for glazing, laminate layers, air gaps, insulation, foil barriers, cavity partitions, curtain walls, and building-integrated antenna structures. ITU-R P.2040 covers basic material properties plus reflection and transmission through materials and multilayer structures. [web:57][web:32]

# Facade archetypes

## Concrete and masonry

Use a frequency-dependent lossy dielectric with dry, nominal, and wet states. Important variables include thickness, water content, aggregate, porosity, render, frequency, and roughness.

## Reinforced concrete

$$L_{RC} = L_{concrete} + L_{rebar} + L_{scatter} + X_{\sigma,RC}$$

| Rebar regime | Condition | Recommended approach |
|---|---|---|
| Dense grid | $s \ll \lambda$ | Conductive screen or high-loss equivalent layer |
| Transitional grid | $s \sim \lambda/2$ to $\lambda$ | Explicit grid, unit cell, or calibrated excess loss |
| Sparse bars | $s \gg \lambda$ | Explicit bars only for critical paths |
| Unknown construction | No rebar data | High-loss class with broad uncertainty |

At mmWave, rebar can be electrically significant and create polarization sensitivity, localized leakage, diffraction, and spatially variable indoor coverage.

### Glass and coated glazing

Model plain glass as a finite dielectric slab; model double glazing, laminated glass, and cavities as multilayer stacks. Low-E, IRR, and solar-control glazing may have conductive coatings that dominate RF interaction.

$$Z_s \approx 1/(\sigma_{ct} c), Y_s = 1/Z_s$$

$$\Gamma_{sheet} = -\eta_0 Y_s/(2 + \eta_0 Y_s), T_{sheet} = 2/(2 + \eta_0 Y_s)$$

Place conductive sheets at their actual position in the glazing stack. Do not classify all glass as ordinary glass.

### Metal cladding and foil

$$\delta = \sqrt{[2/(\omega \mu \sigma)]}$$

If panel thickness is roughly three skin depths or more, treat continuous metal as an opaque reflector. Model seams, vents, apertures, doors, cable entries, and edge diffraction as the realistic leakage routes.

## Roughness and diffuse scattering

$$\sigma_h < \lambda/(8 \cos \theta_i)$$

$$\rho_{coh} = \exp\{-\tfrac{1}{2}[4\pi \sigma_h \cos\theta_i/\lambda]^2\}, \Gamma_{coh} = \rho_{coh} \Gamma_{smooth}$$

$$P_{spec} = |\Gamma_{coh}|^2 P_i$$

$$P_{diff} = \zeta[|\Gamma_{smooth}|^2 - |\Gamma_{coh}|^2] P_i$$

$$p(\theta_s) \propto \cos^m(\theta_s)$$

A facade may be smooth at sub-6 GHz and rough at mmWave. Route coherence-reduced energy consistently to diffuse scatter or unresolved loss; never count it twice. Roughness RMS height and correlation length are key inputs for diffuse-scattering characterisation across 8, 12, and 28 GHz. [web:56]

## Statistical outdoor-to-indoor model

$$PL_{O2I} = PL_{out} + PL_{tw} + PL_{in} + X_{\sigma,O2I}$$

$$PL_{tw} = PL_{angle} - 10 \log_{10}[\ \Sigma_i p_i 10^{(-L_i/10)}\ ]$$

Combine parallel facade pathways in linear transmission space, never by directly averaging dB loss.

$$L_{glass} = 2 + 0.2 f_c; L_{IRRglass} = 25.4 + 0.11 f_c; L_{concrete} = 5 + 4 f_c$$

$$PL_{tw,low} = 5 - 10 \log_{10}[0.3 \cdot 10^{(-L_{glass}/10)} + 0.7 \cdot 10^{(-L_{concrete}/10)}]$$

$$PL_{tw,high} = 5 - 10 \log_{10}[0.7 \cdot 10^{(-L_{IRRglass}/10)} + 0.3 \cdot 10^{(-L_{concrete}/10)}]$$

$$PL_{in} = 0.5 d_{in}$$

| 28 GHz O2I case | Wall term | Indoor term at 10 m | Added median loss |
|---|---|---|---|
| Low-loss façade | ~17.8 dB | 5.0 dB | ~22.8 dB |
| High-loss façade | ~35.0 dB | 5.0 dB | ~40.0 dB |

3GPP TR 38.901 defines O2I building penetration models for 0.5–100 GHz, including low- and high-loss cases. These are system-level abstractions, not universal material constants. [web:59][web:46]

# Measurement, implementation, and QA

$$H(f) = S_{21,material}(f) / S_{21,reference}(f),\ L_{trans}(f) = -20\log_{10}|H(f)|$$

$$R(f) \approx |S_{11}(f)|^2,\ T(f) \approx |S_{21}(f)|^2$$

Use controlled reference measurements with calibration, time gating, and antenna de-embedding. Sample frequency, angle, TE/TM and cross-polar cases, facade position, and dry/wet state.

```
facade_interaction_pipeline:
  - intersect_ray_with_facade_triangle
  - retrieve_material_stack_and_uncertainty_state
  - calculate_incidence_angle_and_local_TE_TM_basis
  - calculate_complex_interface_or_multilayer_coefficients
  - launch_coherent_specular_reflection
  - launch_transmitted_ray_if_above_threshold
  - allocate_roughness_induced_diffuse_power
  - coherently_sum_receiver_fields
  - convert_total_field_to_received_power
  - apply_uncertainty_only_to_unresolved_effects
```

| QA domain | Acceptance checks |
|---|---|
| Physics | TE/TM converge at normal incidence; passive slabs satisfy $R+T \le 1$; no passive gain |
| Numerical | Lossy media decay; matrix is stable; field and power domains are not mixed |
| Geospatial | Normals are correct; windows/frames/walls are separable; layer order is plausible |
| Integrity | No deterministic wall loss plus full empirical O2I wall loss; no double-counted diffuse power |

Report P05, P50, and P95 penetration loss or received power, tagged with frequency, facade class, incidence angle, polarization, and material state.

# References


ITU-R P.2040: Effects of building materials and structures on radiowave propagation in the range of 1 MHz to 450 GHz. [web:57][web:32]

3GPP TR 38.901: Study on channel model for frequencies from 0.5 to 100 GHz, including O2I building penetration loss. [web:59][web:46]

Diffuse-scattering studies at 8, 12, and 28 GHz. [web:56]